

APAR POKHREL

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Education

The University of Texas at Arlington

August 2024 – May 2026

Master of Science in Computer Science, Machine Learning & Computer Vision

GPA: 4.0/4.0

The University of Texas at Arlington

May 2022

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.81/4.0

Work Experience

Research Assistant

May 2025 – Present

PRIMaL Lab | Python, PyTorch, CUDA, NumPy, SciPy, MMGeneration

Arlington, Texas

- Develop and evaluate Diffusion Model-based pipelines for 3-D scene reconstruction using Neural Point Clouds, **targeting 15% improvement in reconstruction fidelity** and **20% faster training convergence** on ShapeNet benchmark.
- Boost model selection efficiency by 30% through literature reviews and ablation experiments on hyperparameter sweeps.

Robotics Research

August 2025 – Present

Robotic Vision Laboratory | Python, PyTorch, ROS/ROS2, OpenCV

Arlington, Texas

- Benchmark Vision-Language-Action models (TinyVLA, SmolVLA) in robotic manipulation tasks and deploy SmolVLA on NVIDIA Jetson, **targeting real-time control with ≤ 30 ms latency** and **95% task success rate** with UR5.

Graduate Teaching Assistant, bugHouse Tutor

August 2024 – Present

The University of Texas at Arlington

Arlington, Texas

- Enhanced learning outcomes for 200+ students** in Object-Oriented Programming (Java) and various lower/upper level CSE courses by providing tutoring, designing supplemental exercises, and supporting faculty-led instruction.

Instructor

May 2025 – July 2025

Upward Bound Math & Science Center

Arlington, Texas

- Directed a project-based curriculum introducing 30+ students to Drone Design and Bioinformatics**, guiding drone assembly, prototypes, flight tests, and mentoring teams on genomic analysis and medical image segmentation.

Software Engineer

September 2022 – February 2024

SAE International | Java, Spring, JavaScript, Oracle, PostgreSQL, AWS

Warrendale, Pennsylvania

- Resolved 50+ production bug tickets**, leveraging AWS CloudWatch and Elasticsearch for log tracing and documented Root Cause Analysis to reduce repeat incidents by 25%.
- Developed and optimized store procedures to streamline audit and expense workflows, **slashing support ticket response time by 75%**, and resolving data exchange failures by dissecting failure codes and server logs.
- Engineered a serverless PDF reporting pipeline on AWS Lambda with SendGrid API, automating report generation and email delivery, and real-time notifications, **cutting distribution time by 60%**.
- Integrated Keycloak for user auth and SSO with **role-based access for 1,000+ users**, implementing OIDC/OAuth2 token flows and automated provisioning, and documented API endpoints in Swagger UI for seamless integration.

Student Developer, IEEE Robotics

August 2021 – May 2022

The University of Texas at Arlington | Python, C++, Raspberry Pi, ReactJS

Arlington, Texas

- Collaborated on designs and builds for an underwater ROV with a \$800 budget for the IEEE Robotics Competition.
- Developed an interface with live video streaming to enable real-time maneuvers, reducing operator response time by 30%.

Technical Skills

Languages: Python, Java, JavaScript, SQL, C/C++

Database & APIs: Oracle, MySQL, PostgreSQL, MongoDB, OpenAPI, FastAPI, Swagger, REST

Libraries & Frameworks : ReactJS, Axios, Jest, NodeJS, ExpressJS, Bootstrap, TailwindCSS

Cloud & Dev Tools: AWS (EC2, S3, Lambda), GCP, Git, Gitlab CI/CD, Docker, Jira, Postman, ELTK Stack

Machine Learning: PyTorch, scikit-learn, NumPy, Pandas, NLTK

Projects

Parking Occupancy Detection – PKLot | Python, PyTorch

May 2025

- [Paper] Evaluated YOLOv8 backbones (ResNet-18, VGG16, EfficientNet, Ghost-P2), achieving up to 98.6% mAP50:95, 0.9–4.1ms inference latency, 47% reduction in model parameters with computational efficiency as low as 8.2 GFLOPs.

Hackathons | Flask, ReactJS, Bootstrap, Axios, Python, Tensorflow

September 2022

- Insure It:** Tesseract OCR- based app to parse consumer receipts for insurance quotes. (**1st place@HackUTA 2022**)
- CareHive** (3rd place@Hackdemonium 2.0 2022), **AgroHive** (HackTX 2022), **10 years with Jake** (HackDFW 2022)